Homogenous Coordinates

I Homogenous Transforms

- Trick → Append 1 to the end of the point

$${}_1q = {}_1P + {}_1R\,{}_2q$$

$${}_1\bar{q} = \begin{bmatrix} {}_1R & {}_1P \\ 0 & 1 \end{bmatrix} {}_2\bar{q} = \begin{bmatrix} {}_1R & {}_1P \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}_2q \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} {}_1R\,{}_2q + {}_1P \\ 1 \end{bmatrix} = {}_1T\,{}_2\bar{q}$$

$SE(n)$ Group $(n=2,3)$

Identity: $T \cdot I = T$

Inverse: ${}_1T^{-1} = {}_2{}_1T$, ${}_2{}_1T = \begin{bmatrix} {}_1R^T & -{}_1R^T\,{}_1P \\ 0 & 1 \end{bmatrix}$

Group Operation: $(T_1 \circ T_2) \in SE(n)$ ← Matrix Multiplication

Closure: $T_A \cdot T_B \in SE(3)$

$$\Rightarrow = \begin{bmatrix} R_A & P_A \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} R_B & R_B \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \overbrace{R_A R_B}^{\in SO(3)} & \overbrace{R_A P_B + P_A}^{\in \mathbb{R}^3} \\ 0 & 1 \end{bmatrix}$$

II Chaining Transforms

Suppose we have 3 frames A, B, C, and W where W is our world coordinate.

Also suppose we know their transforms, i.e.

$$W \xleftarrow[T_{WB}]{} B \xrightarrow[T_{CB}]{} C \xleftarrow[T_{CA}]{} A$$

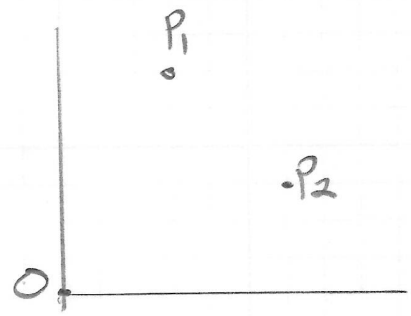
where C is our robot and A, B are data represented in another coordinate

Suppose we have ${}_Aq$, what is ${}_Wq$.

To make ${}_Aq$ into a homogenous coordinate ${}_A\bar{q} = \begin{bmatrix} {}_Aq \\ 1 \end{bmatrix}$

$\Rightarrow$ $\underline{{}_W\bar{q} = {}_CT\,{}_BT^{-1}\,{}_WT\,{}_A\bar{q}}$ where ${}_iT^{-1}$ is defined above

III Homogenous Properties



- Difference between vector and point
  - Point: coordinate or position
  - Vector: Direction & magnitude
- Standard Coordinate: No difference

$$P_1 = \begin{bmatrix} P_{11} \\ P_{12} \\ P_{13} \\ \vdots \end{bmatrix} \quad P_2 = \begin{bmatrix} P_{21} \\ P_{22} \\ P_{23} \\ \vdots \end{bmatrix} \quad \vec{P}_1 = \begin{bmatrix} P_{11} \\ P_{12} \\ P_{13} \\ \vdots \end{bmatrix} \quad \vec{P}_2 = \begin{bmatrix} P_{21} \\ P_{22} \\ P_{23} \\ \vdots \end{bmatrix}$$

Same Representation

- Homogenous

$$\bar{P}_1 = \begin{bmatrix} P_{11} \\ P_{12} \\ P_{13} \\ \vdots \\ 1 \end{bmatrix} \quad \bar{P}_2 = \begin{bmatrix} P_{21} \\ P_{22} \\ P_{23} \\ \vdots \\ 1 \end{bmatrix} \quad \bar{O} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

Points have a 1 at the end of them

$$\vec{\bar{P}}_1 = \vec{\bar{P}}_1 - \vec{\bar{q}} = \begin{bmatrix} P_{11} \\ P_{12} \\ P_{13} \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} P_{11} \\ P_{12} \\ P_{13} \\ 0 \end{bmatrix}$$

Vectors have a 0 at the end of them

A) Vector side note

- Vectors always start at the origin of any frame and has direction and magnitude.

B) Properties Summary

- Sum & difference of vectors: are vectors
- Sum of a vector and point is a point
- Difference between two points is a vector
- Sum of two points has no meaning

=> from the properties defined

- Transforming vectors we want no translation. Rotation only
- Transforming points we want both rotation and translation

c) Standard rotation Transform

$$ {}^{A}B_T = \begin{bmatrix} {}^{A}B_R & {}^{A}B_P \\ 0 & 1 \end{bmatrix} $$

Lets transform a point

$$ {}^{A}\bar{q} = \begin{bmatrix} {}^{A}B_R & {}^{A}B_P \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^{B}q \\ 1 \end{bmatrix} = \begin{bmatrix} {}^{A}B_R\,{}^{B}q + {}^{A}B_P \\ 1 \end{bmatrix} $$

Lets transform a vector

$$ {}^{A}\bar{v} = \begin{bmatrix} {}^{A}B_R & {}^{A}B_P \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^{B}v \\ 0 \end{bmatrix} = \begin{bmatrix} {}^{A}B_R\,{}^{B}v + {}^{A}B_P \cdot 0 \\ 0\,{}^{B}v + 0 \cdot 1 \end{bmatrix} $$

$$ = {}^{A}B_R\,{}^{B}v $$